

Infosys Springboard Virtual Internship 6.0 Completion Report

Team Details

Batch Number 13

Start date : 02-Feb-2026

Name: Vishok P

Internship Duration: 8 Weeks

1. Project Title

Infosys_InsurAI – Corporate Policy Automation & Intelligence System

2. Project Objective

The main objective of this project was to develop a secure, AI-assisted web platform that automates end-to-end customer interaction and appointment workflows for a corporate insurance environment by integrating user authentication and registration, voice-based query handling, appointment scheduling, agent availability management, plan information management, appointment tracking, notification services, and a admin dashboard into a single cohesive system, so that customers can easily get information and book services online, agent can manage their availability efficiently, and administrators can monitor all the activities centrally- thereby reducing manual work improving operational efficiency, and enhancing the overall customer experience in line with an automation and AI-driven insurance strategy.

3. Project description in detail

The Infosys_InsurAI – Corporate Policy Automation & Intelligence System internship project is a full-stack insurance platform that digitizes user on-boarding, secure access, and core policy interactions using a layered architecture of React.js, Spring Boot, MySQL, and JWT-based authentication. It is designed so that customers, agents, and admins can securely log in, manage policies, schedule appointments, and validate business rules through a robust API-driven back-end and a responsive front-end, similar to real production-grade insurance and service applications.

Overall approach:

The project was structured using a clear separation of concerns: the front-end handled all user interaction, while the back-end exposed REST API s for business logic and database operations. React.js was used to build reusable UI components for login, registration, dashboards, plan views, and appointment screens, and these components communicated with Spring Boot API s via HTTP calls (typically using fetch or Axios). The back-end defined controllers, services, and repositories; controllers received requests from the UI, services implemented business rules

(such as validating appointment slots or access rights), and repositories used JPA/Hibernate to interact with MySQL for persistent storage of users, policies, agents, and appointments.

Technology stack and layer roles:

At the front-end layer, React.js provided routing between pages (login, user dashboard, admin dashboard, plan details, appointment screens) and managed state for forms, API responses, and notifications, enabling a SPA-like experience with minimal page reloads. The back-end used Spring Boot with Spring Web and Spring Security to expose Restful endpoints, encapsulate business logic, and enforce authorization checks before returning or modifying data, fitting well with enterprise insurance requirements. MySQL served as the relational database to store structured entities such as users, roles, policies, plans, appointments, and notifications; this choice matched typical enterprise insurance systems where strong data integrity and relational queries are needed.

Security and authentication with JWT:

Authentication was implemented using JWT (JSON Web Token) integrated with Spring Security: when a user logged in with valid credentials, the back-end validated the user against MySQL, generated a signed JWT, and sent it back to the client. The React front-end stored the token (for example in memory or secure storage) and attached it in the Authorization header for each subsequent API call, allowing the back-end to verify the token, extract user identity and roles, and authorize access to protected endpoints such as admin functions or sensitive policy data. This token-based approach made the system stateless and scalable, which is aligned with how modern insurance platforms handle authentication for web and mobile clients.

Testing and validation:

For testing the back-end APIs, tools like Postman were used to verify endpoints for login, registration, policy retrieval, appointment booking, and admin operations, checking both success paths and error scenarios (unauthorized, validation errors, missing data). Browser developer tools were then used to debug front-end–back-end integration, inspect network requests, monitor JWT headers, and ensure that protected routes were correctly blocked or allowed based on login status and role. This combination of API-level testing and UI-level validation helped ensure that the system behaved reliably and consistently across different user flows.

Real-world impact and relevance:

In a real insurance setting, a platform like Infosys_InsurAI can significantly reduce manual work by automating policy and appointment workflows, enabling self-service for customers and structured dashboards for agents and admins. Real implementations of similar stacks in insurance and related domains have shown faster processing of customer requests, reduced operational costs, and better compliance and audit trails due to centralized, structured data. Using mainstream technologies (React, Spring Boot, MySQL, JWT) also means this project is directly transferable to enterprise environments where Infosys and insurers are already building AI- and automation-driven systems, making the internship work relevant as a blueprint or prototype for production solutions.

4. Timeline Overview

Week	Activities Planned	Activities Completed
Week 1	Project Abstraction, Setup for React	Completed React application structure, routing setup, component architecture
Week 2	Implement the Authentication System	User login and register page is created
Week 3	Developed policy management in the project	Created information about the insurances that are implemented in the project
Week 4	Agent managaement	Created agent that is to connect the user and company.
Week 5	Appointment Booking Management	Created appointment booking system where user user can book particular time, insurance and the agent.
Week 6	To create Dashboard and user module	User dashboard with activity overview, appointment history.
Week 7	Notification system implementation	Created notification bar for the project
Week 8	To create admin module	Admin module is created to monitor the user, agent and policy details.

5a. Key Milestones

Milestone	Description	Status
Project Kickoff	Defined project scope, objectives, features, and finalized tech stack (React.js, Spring Boot, MySQL, JWT) with mentor approval.	Completed
Prototype/First Draft	Completed first working prototype with basic authentication, policy listing, and initial appointment booking flow.	Completed
Mid-Term Review	Demonstrated core modules to mentor (login/registration, policies, appointments), received feedback, and refined the development plan.	Completed
Final Submission	Delivered the complete Infosys_InsurAI system with dashboards, notifications, role-based access, testing, and documentation.	Completed
Presentation	Presented the project, architecture, features, and real-world insurance	Completed

	relevance to the evaluation panel.	
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5b. Project execution details

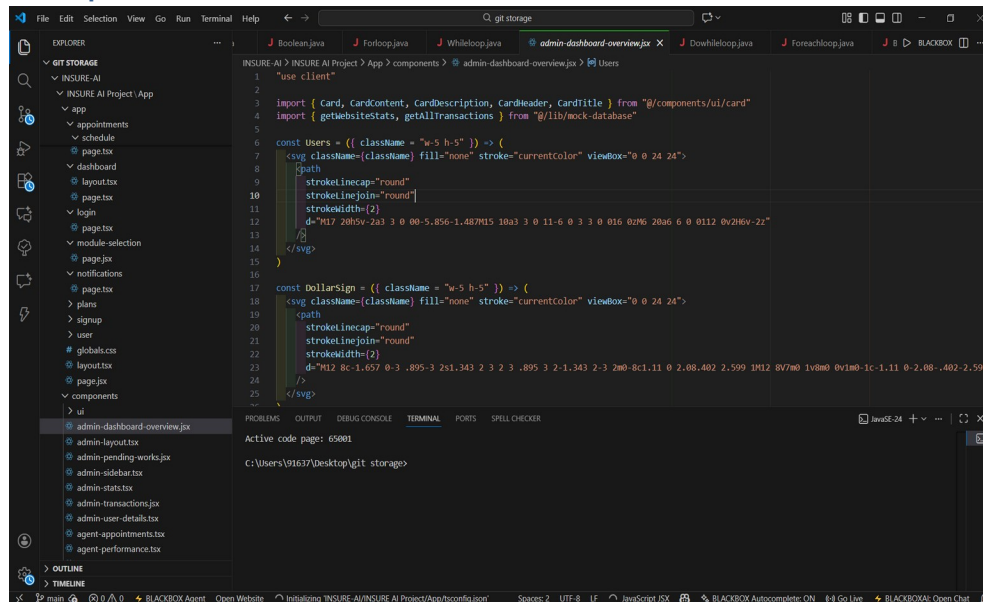
This project was executed in planned phases aligned with clear milestones over the internship timeline.

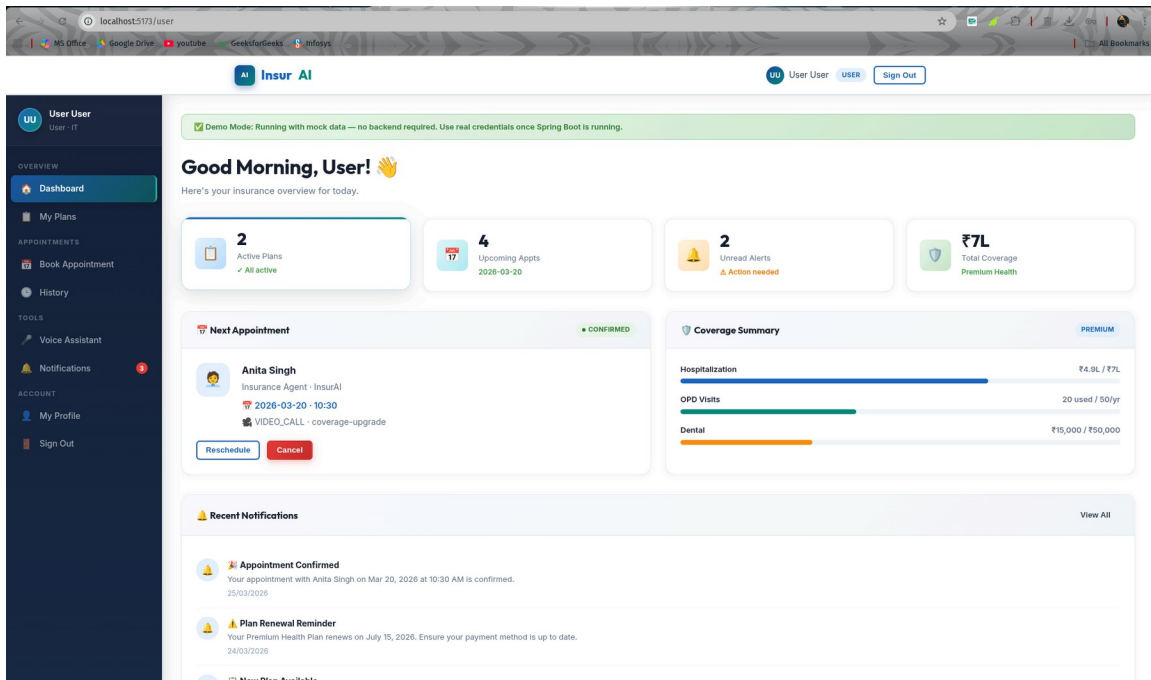
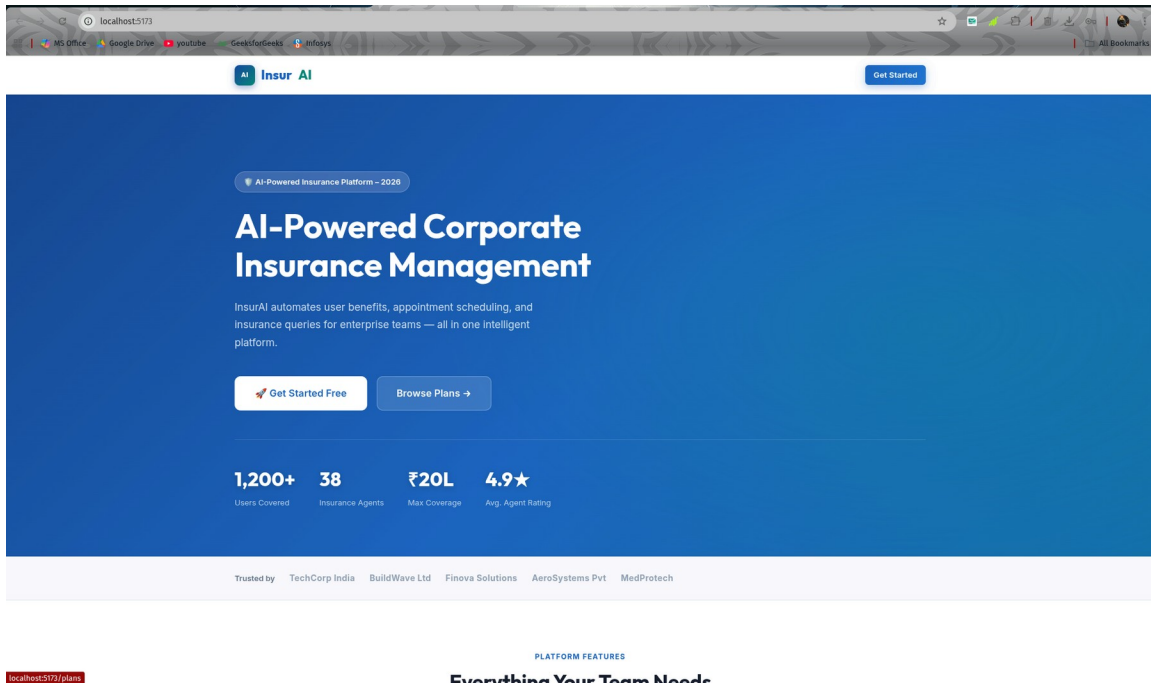
The work started with a kickoff where scope, features, and the tech stack (React.js, Spring Boot, MySQL, JWT) were finalized and the week-wise plan was defined. A basic full-stack skeleton was then created: React for UI, Spring Boot for APIs, and MySQL for core entities like users, policies, and appointments, forming the first working prototype.

Next, core features were implemented: JWT authentication and role-based access, policy listing and details, appointment booking, dashboards, and notification components, all wired to secured Spring Boot endpoints. After the mid-term review, UI/UX and flows were refined and thoroughly tested using Postman for APIs and browser tools for front-end-back-end integration and JWT handling.

In the final phase, modules were integrated into a stable Infosys_InsurAI system, code was cleaned, documentation was written, and the project was prepared as a deployable prototype. The internship concluded with final submission and a presentation highlighting architecture, tech choices, features, and real-world relevance for automated insurance workflows.

6. Snapshots / Screenshots





localhost:5173/agent

AS Anita Singh AGENT Sign Out

Good Morning, Anita! 🌞

Your performance overview for today.

2
Pending Requests
Action required

1
Today's Meetings
All confirmed

28
Active Customers
3 new this month

4.9
Avg. Rating
87 reviews

Today's Schedule

MARCH 8

10:30 Ravi Kumar - coverage-upgrade
VIDEO_CALL CONFIRMED

Monthly Performance

Oct Nov Dec Jan Feb Mar

87% Acceptance Rate

24 Appts This Month

Welcome! 🌞
Logged in as Agent

SA Super Admin ADMIN Sign Out

Admin Analytics Dashboard

Real-time overview of the entire InsurAI platform.

1,247
Total Users
+42 this month

38
Active Agents
+3 this month

284
Appointments (MTD)
+18% vs last month

8
Active Plans
+2 new plans

Monthly Appointment Trend

2025-26

Sep Oct Nov Dec Jan Feb Mar

Plan Distribution

Basic Health Enterprise Group Life Others Premium Health

Recent Appointments

USER	AGENT	DATE	STATUS
Ravi Kumar	Anita Singh	Mar 9	CONFIRMED
Meena Patel	Anita Singh	Mar 9	CONFIRMED
Arjun Joshi	Rajesh Mehta	Mar 10	PENDING
Nisha Gupta	Priya Nair	Mar 11	RESCHEDULED

System Alerts

6 Plans Expiring
6 users have plans expiring within 30 days.

System Healthy
All services running normally. Uptime: 99.97%

New Agent Applications
3 new agent applications pending review.

7. Challenges Faced

Throughout the internship, I faced several technical and non-technical challenges:

Technical Challenges:

- Integrating React.js with Spring Boot APIs initially caused CORS and response format issues, fixed through proper CORS setup and consistent JSON structures.
- Implementing JWT authentication and role-based access was complex, mitigated by following Spring Security + JWT patterns and testing each secured endpoint.
- Designing a scalable MySQL schema for users, policies, appointments, and notifications was challenging and resolved with iterative normalization and clear relationships.
- Handling validation and error messages across frontend and backend was confusing at first, improved by standardizing error formats and mapping them cleanly in the UI.
- Debugging full-stack integration issues took time and was eased using Postman, browser dev tools, and stepwise testing of each module.

Non-Technical Challenges:

- Balancing weekly development targets with bug fixing was difficult, managed by prioritizing core features and splitting work into smaller tasks
- Unclear initial expectations for features like dashboards and notifications caused rework, reduced through regular mentor discussions and early requirement confirmation.
- Frontend-backend contract changes created integration mismatches, mitigated by documenting APIs and communicating changes before implementation.
- Managing time around milestones (prototype, mid-term, final submission, presentation) was challenging, improved by tracking progress against a simple timeline.
- Explaining technical decisions to non-technical stakeholders was hard initially, addressed by using simple diagrams and insurance-based examples in reviews and presentations.

8. Learnings & Skills Acquired

Technical Skill:

- React.js component development, hooks, and SPA UI building.
- Spring Boot REST API development with layered architecture (controller-service-repository).
- Spring Security with JWT authentication and role-based authorization.
- MySQL database design, schema creation, and CRUD operations for users, policies, and appointments.
- API testing with Postman and frontend integration debugging using browser developer tools.

Soft Skills:

- Problem-solving across frontend, backend, and database integration issues.
- Time management using weekly plans, milestones, and deadlines (prototype, review, final).
- Project planning and feature sequencing from core flows to advanced enhancements.
- Requirement clarification and communication through regular mentor discussions and demos.
- Documentation and presentation of architecture, features, and real-world impact of the project.

9. Testimonials from team

During this internship, the Infosys_InsurAI project was taken from idea to a working full-stack prototype with secure authentication, policy and appointment workflows, and role-based dashboards using React.js, Spring Boot, MySQL, and JWT. The milestones were delivered on time, issues across UI, APIs, and database were systematically debugged using Postman and browser tools, and each review showed clear improvement in both functionality and usability. Overall, the experience strengthened practical technical skills, problem-solving, and communication, and resulted in a project that closely reflects real-world automated insurance solutions.

10. Conclusion

The Infosys_InsurAI – Corporate Policy Automation & Intelligence System internship provided end-to-end exposure to building a real-world style full-stack solution using React.js, Spring Boot, MySQL, and JWT, closely mirroring the technologies and patterns used in modern enterprise insurance platforms. Working through clearly defined milestones—from project kickoff and prototype to final submission and presentation—improved technical depth, structured thinking, and confidence in handling secure authentication, REST APIs, data modeling, and UI integration. The project's focus on automating policy, appointment, and notification workflows directly connects classroom concepts of web development, databases, and software engineering with practical industry use cases, making it a strong foundation for a career in full-stack and insurance/Fin-tech-related development.

11. Acknowledgements

This internship would not have been possible without the guidance and support of the organization and mentors involved. Sincere thanks to Infosys for providing the opportunity to work on the Infosys_InsurAI – Corporate Policy Automation & Intelligence System, which offered real exposure to industry-relevant technologies and insurance workflows. Special thanks to the mentor and team members for their continuous feedback, technical guidance, and encouragement throughout the project milestones, from initial planning to final presentation. Their support played a key role in building both technical skills and professional confidence during this internship journey.